

```
import pandas as pd
import matplotlib.pyplot as plt
```

```
from google.colab import files
```

```
uploaded = files.upload()
```

No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.
Saving sales_data.csv to sales_data.csv

```
df = pd.read_csv("sales_data.csv")
```

```
df.head()
```

	Product_ID	Sale_Date	Sales_Rep	Region	Sales_Amount	Quantity_Sold	Product_Category
0	1052	2023-02-03	Bob	North	5053.97	18	Furniture
1	1093	2023-04-21	Bob	West	4384.02	17	Furniture
2	1015	2023-09-21	David	South	4631.23	30	Food
3	1072	2023-08-24	Bob	South	2167.94	39	Clothing
4	1061	2023-03-24	Charlie	East	3750.20	13	Electronics

```
df.tail()
```

	Product_ID	Sale_Date	Sales_Rep	Region	Sales_Amount	Quantity_Sold	Product_Category
995	1010	2023-04-15	Charlie	North	4733.88	4	Food
996	1067	2023-09-07	Bob	North	4716.36	37	Clothing
997	1018	2023-04-27	David	South	7629.70	17	Clothing
998	1100	2023-12-20	David	West	1629.47	39	Electronics
999	1086	2023-08-16	Alice	East	4923.93	48	Food

```
df.shape
```

```
(1000, 14)
```

```
df.columns
```

```
Index(['Product_ID', 'Sale_Date', 'Sales_Rep', 'Region', 'Sales_Amount',  
      'Quantity_Sold', 'Product_Category', 'Unit_Cost', 'Unit_Price',  
      'Customer_Type', 'Discount', 'Payment_Method', 'Sales_Channel',  
      'Region_and_Sales_Rep'],  
      dtype='object')
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 14 columns):
#   Column                                Non-Null Count  Dtype  
---  -
0   Product_ID                           1000 non-null   int64   
1   Sale_Date                             1000 non-null   object  
2   Sales_Rep                             1000 non-null   object  
3   Region                                1000 non-null   object  
4   Sales_Amount                          1000 non-null   float64  
5   Quantity_Sold                         1000 non-null   int64   
6   Product_Category                      1000 non-null   object  
7   Unit_Cost                             1000 non-null   float64  
8   Unit_Price                           1000 non-null   float64  
9   Customer_Type                         1000 non-null   object  
10  Discount                              1000 non-null   float64  
11  Payment_Method                        1000 non-null   object  
12  Sales_Channel                         1000 non-null   object  
13  Region_and_Sales_Rep                  1000 non-null   object  
dtypes: float64(4), int64(2), object(8)
memory usage: 109.5+ KB
```

```
df.describe()
```

	Product_ID	Sales_Amount	Quantity_Sold	Unit_Cost	Unit_Price	Discount
count	1000.000000	1000.000000	1000.000000	1000.000000	1000.000000	1000.000000
mean	1050.128000	5019.265230	25.355000	2475.304550	2728.440120	0.15239
std	29.573505	2846.790126	14.159006	1417.872546	1419.399839	0.08720
min	1001.000000	100.120000	1.000000	60.280000	167.120000	0.00000
25%	1024.000000	2550.297500	13.000000	1238.380000	1509.085000	0.08000
50%	1051.000000	5019.300000	25.000000	2467.235000	2696.400000	0.15000
75%	1075.000000	7507.445000	38.000000	3702.865000	3957.970000	0.23000
max	1100.000000	9989.040000	49.000000	4995.300000	5442.150000	0.30000

```
df.isnull().sum()
```

	0
Product_ID	0
Sale_Date	0
Sales_Rep	0
Region	0
Sales_Amount	0
Quantity_Sold	0
Product_Category	0
Unit_Cost	0
Unit_Price	0
Customer_Type	0
Discount	0
Payment_Method	0
Sales_Channel	0
Region_and_Sales_Rep	0

dtype: int64

```
df = df.dropna()
```

```
df.isnull().sum()
```

	0
Product_ID	0
Sale_Date	0
Sales_Rep	0
Region	0
Sales_Amount	0
Quantity_Sold	0
Product_Category	0
Unit_Cost	0
Unit_Price	0
Customer_Type	0
Discount	0
Payment_Method	0
Sales_Channel	0
Region_and_Sales_Rep	0

dtype: int64

```
df.duplicated().sum()
```

```
np.int64(0)
```

```
df = df.drop_duplicates()
```

```
df.duplicated().sum()
```

```
np.int64(0)
```

```
df.nunique()
```

	0
Product_ID	100
Sale_Date	340
Sales_Rep	5
Region	4
Sales_Amount	998
Quantity_Sold	49
Product_Category	4
Unit_Cost	998
Unit_Price	1000
Customer_Type	2
Discount	31
Payment_Method	3
Sales_Channel	2
Region_and_Sales_Rep	20

dtype: int64

```
df["Sales_Amount"].sum()
```

```
np.float64(5019265.2299999995)
```

```
df["Sales_Amount"].mean()
```

```
np.float64(5019.265229999999)
```

```
df.groupby("Region")["Sales_Amount"].sum()
```

	Sales_Amount
Region	
East	1259792.93
North	1369612.51
South	1154250.86
West	1235608.93

dtype: float64

```
df.groupby("Product_Category")["Sales_Amount"].sum()
```

```
df.groupby("Product_Category")["Sales_Amount"]
```

Product_Category	Sales_Amount
Clothing	1313474.36
Electronics	1243499.64
Food	1201773.54
Furniture	1260517.69

dtype: float64

```
df.groupby("Sales_Rep")["Sales_Amount"].sum().sort_values(ascending=False)
```

```
df.groupby("Sales_Rep")["Sales_Amount"]
```

Sales_Rep	Sales_Amount
David	1141737.36
Bob	1080990.63
Eve	970183.99
Alice	965541.77
Charlie	860811.48

dtype: float64

```
df["Discount"].mean()
```

```
np.float64(0.15239)
```

```
df.groupby("Product_Category")["Quantity_Sold"].sum()
```

```
df.groupby("Product_Category")["Quantity_Sold"]
```

Product_Category	Quantity_Sold
Clothing	6922
Electronics	6096
Food	5608
Furniture	6729

dtype: int64

```
df["Payment_Method"].value_counts()
```

	count
Payment_Method	
Credit Card	345
Bank Transfer	342
Cash	313

dtype: int64

```
df["Sales_Channel"].value_counts()
```

	count
Sales_Channel	
Retail	512
Online	488

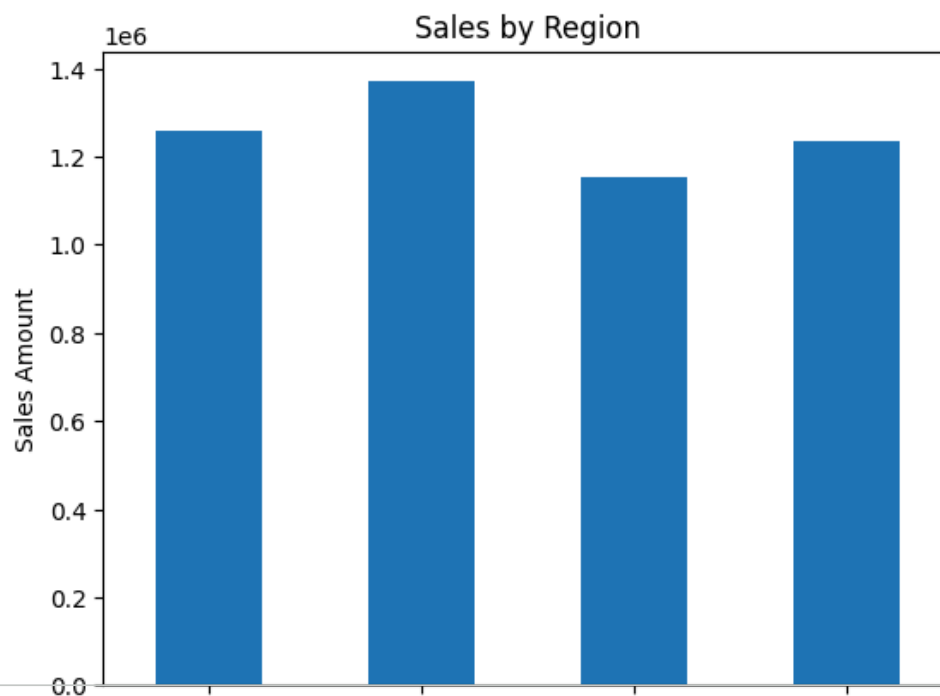
dtype: int64

```
region_sales = df.groupby("Region")["Sales_Amount"].sum()

region_sales.plot(kind="bar")

plt.title("Sales by Region")
plt.xlabel("Region")
plt.ylabel("Sales Amount")

plt.show()
```

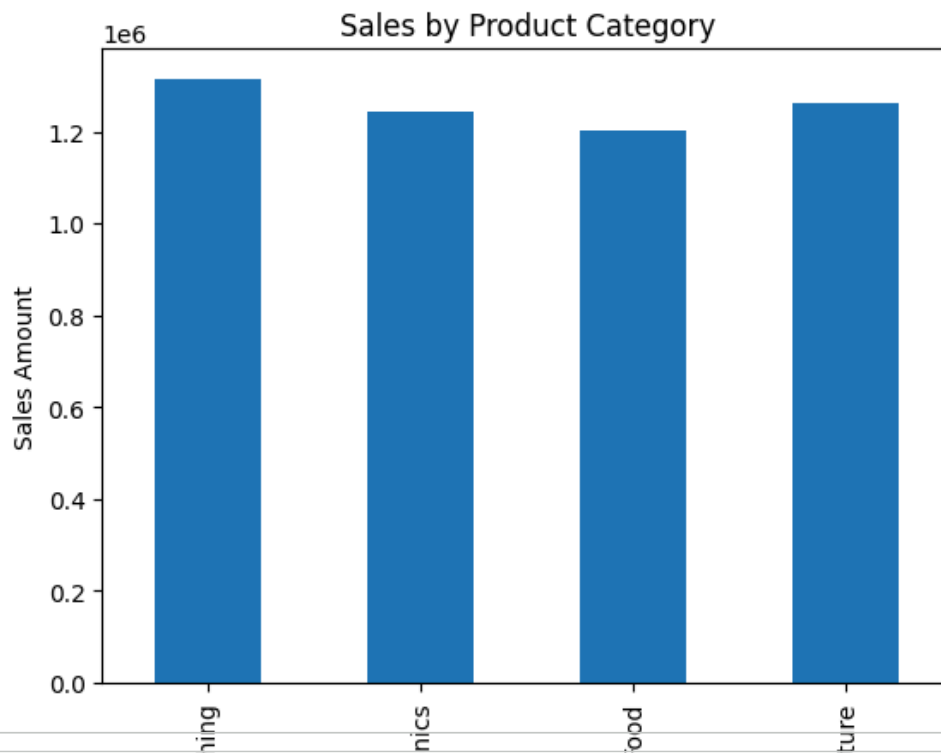


```
category_sales = df.groupby("Product_Category")["Sales_Amount"].sum()

category_sales.plot(kind="bar")
```

```
plt.title("Sales by Product Category")
plt.xlabel("Category")
plt.ylabel("Sales Amount")

plt.show()
```

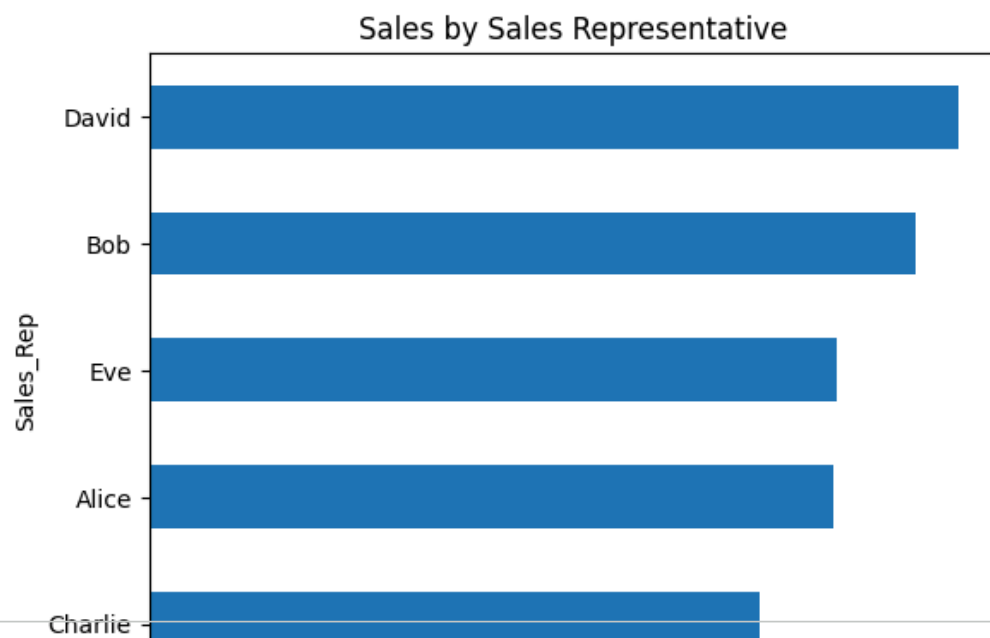


```
rep_sales = df.groupby("Sales_Rep")["Sales_Amount"].sum().sort_values()

rep_sales.plot(kind="barh")

plt.title("Sales by Sales Representative")
plt.xlabel("Sales Amount")

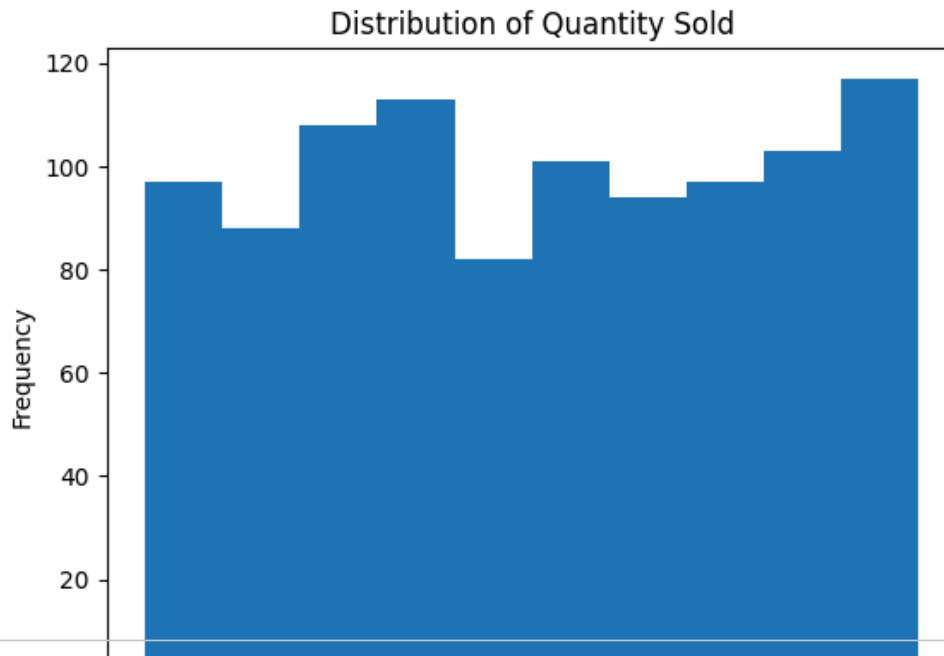
plt.show()
```



```
plt.hist(df["Quantity_Sold"], bins=10)

plt.title("Distribution of Quantity Sold")
plt.xlabel("Quantity Sold")
plt.ylabel("Frequency")

plt.show()
```



```
df["Sale_Date"] = pd.to_datetime(df["Sale_Date"])
```

Quantity Sold

```
df["Month"] = df["Sale_Date"].dt.month_name()
```

```
df[["Sale_Date", "Month"]].head()
```

	Sale_Date	Month
0	2023-02-03	February
1	2023-04-21	April
2	2023-09-21	September
3	2023-08-24	August
4	2023-03-24	March

```
monthly_sales = df.groupby("Month")["Sales_Amount"].sum()

monthly_sales
```

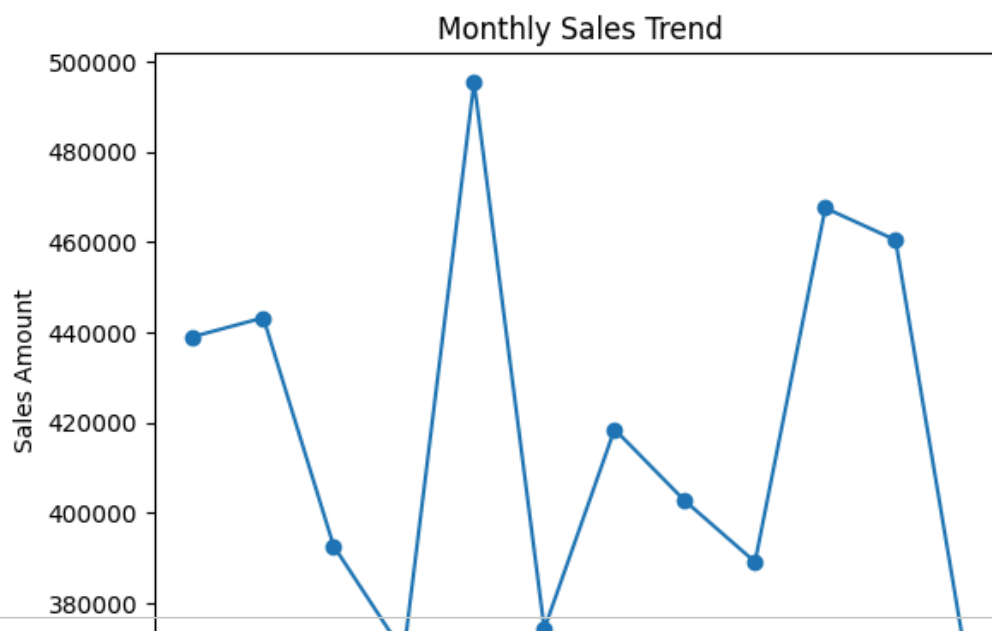
Sales_Amount	
Month	
April	438992.61
August	443171.28
December	392643.58
February	368919.36
January	495420.37
July	374242.88
June	418458.34
March	402638.77
May	389078.76
November	467482.90
October	460378.78
September	367837.60

dtype: float64

```
monthly_sales.plot(kind="line", marker="o")
```

```
plt.title("Monthly Sales Trend")
plt.xlabel("Month")
plt.ylabel("Sales Amount")
```

```
plt.show()
```



```
df.groupby("Customer_Type")["Sales_Amount"].sum()
```

April December January June May October
Month

Sales_Amount	
Customer_Type	
New	2506258.30
Returning	2513006.93

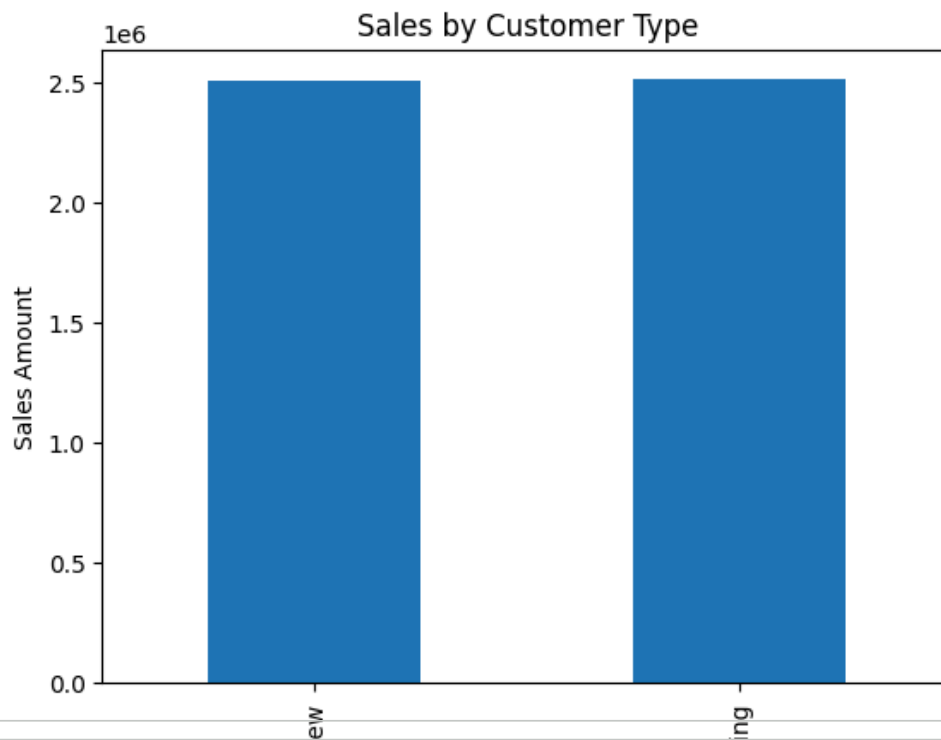
dtype: float64

```
customer_sales = df.groupby("Customer_Type")["Sales_Amount"].sum()

customer_sales.plot(kind="bar")

plt.title("Sales by Customer Type")
plt.xlabel("Customer Type")
plt.ylabel("Sales Amount")

plt.show()
```



```
df.groupby("Product_Category")["Unit_Price"].mean()
```

Unit_Price		Customer Type
Product_Category		
Clothing	2721.256381	
Electronics	2795.207195	
Food	2657.811726	
Furniture	2734.065346	

dtype: float64

```
df.corr(numeric_only=True)
```

	Product_ID	Sales_Amount	Quantity_Sold	Unit_Cost	Unit_Price	Discount
Product_ID	1.000000	0.010221	0.094479	0.013402	0.015116	-0.022097
Sales_Amount	0.010221	1.000000	-0.041599	0.009770	0.011812	0.023153
Quantity_Sold	0.094479	-0.041599	1.000000	0.053888	0.057296	-0.007806
Unit_Cost	0.013402	0.009770	0.053888	1.000000	0.995056	-0.017741
Unit_Price	0.015116	0.011812	0.057296	0.995056	1.000000	-0.017027
Discount	-0.022097	0.023153	-0.007806	-0.017741	-0.017027	1.000000

```
df["Profit_Per_Unit"] = df["Unit_Price"] - df["Unit_Cost"]
```